Co1 At2

SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

GOVINDARAJU VISHNU MADHAV
192425192A

Assignment_02_CO_01

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QUESTIONS

- Q1 Print a hollow diamond pattern — 5 marks
- Q2 Design a class to calculate parking charges — 5 marks
- Q3 Print a number pattern with consecutive values — 5 marks
- Q4 Check whether a number is a spy number — 5 marks
- Q5 Print a butterfly pattern — 5 marks

5/5 answered

Q1 Print a hollow diamond pattern — 5 marks

Use nested loops to print a diamond of five rows in the upper half and four rows in the lower half.

```
*
* *
* *
* *
* *
* *
* *
*
```

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "*" << endl;
    for (int i = 0; i < 7; i++) {
        cout << "* *" << endl;
    }
    cout << "*" << endl;
    return 0;
}
```

Input

stdin input

Output

```
*
* *
* *
* *
* *
* *
* *
*
```

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5/5 answered

Q2 Design a class to calculate parking charges — 5 marks

Create a Parking class. Charge ₹30 for the first hour and ₹20 for every additional hour. Any fraction of an hour counts as a full hour.
TC_1: 2 hours and 15 minutes →Parking charge: Rs.70. TC_2: 0 hours and 30 minutes →Parking charge: Rs.30 .TC_3: 2 hours and 60 minutes → Invalid duration

Your Code

```cpp
#include <iostream>
using namespace std;

class Parking {
public:
    void calculate(int h, int m) {
        if (m >= 60) {
            cout << "Invalid duration" << endl;
            return;
        }
        if (m > 0) h++;
        int charge = 30;
        if (h > 1) charge += (h - 1) * 20;
        cout << "Parking charge: Rs." << charge << endl;
    }
};

int main() {
    Parking p;
    int h, m;
    while (cin >> h >> m) {
        p.calculate(h, m);
    }
    return 0;
}
```

Input

```
2 15
0 30
2 60
```

Output

```
Parking charge: Rs.70
Parking charge: Rs.30
Invalid duration
```

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

5.00 / 5 marks

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- Q4 Check whether a number is a spy number — 5 marks
- Q5 Print a butterfly pattern — 5 marks

5/5 answered

Q3 Print a number pattern with consecutive values — 5 marks

Print the following pattern for four rows.

```
1
2 3
4 5 6
7 8 9 10
```

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 4;
    int num = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << num << " ";
            num++;
        }
        cout << endl;
    }
    return 0;
}
```

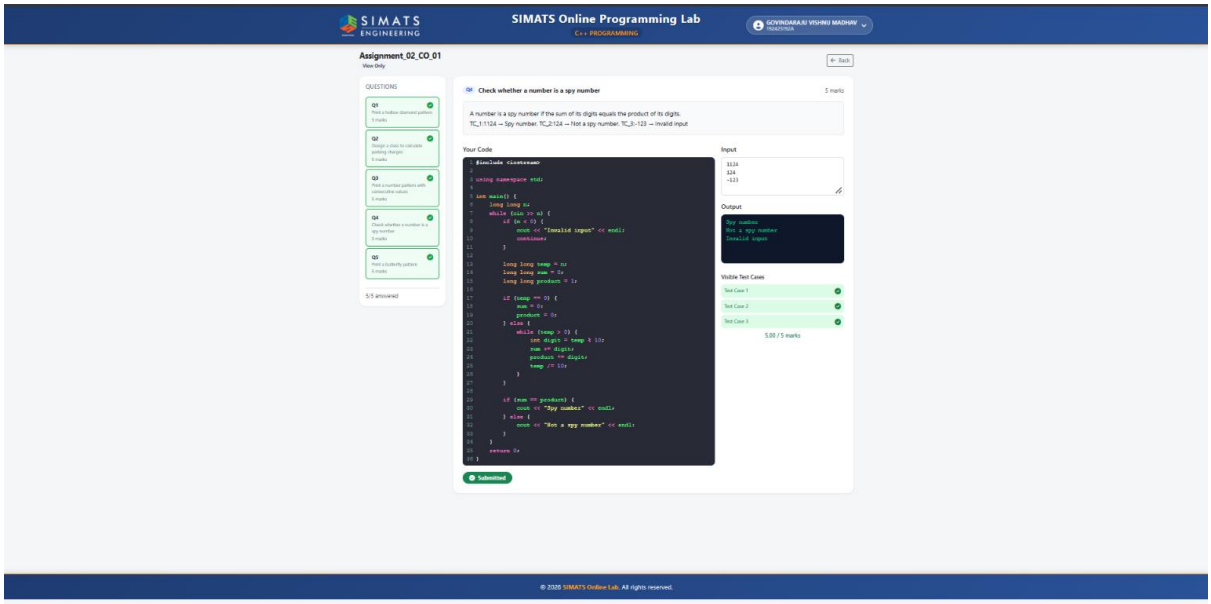
Input

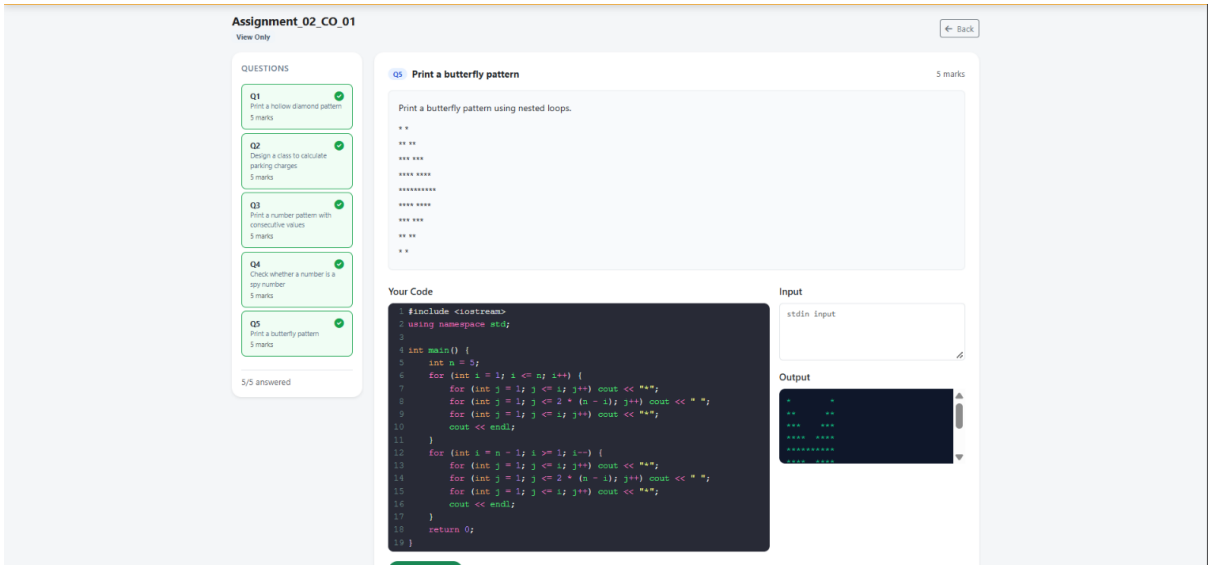
stdin input

Output

```
1
2 3
4 5 6
7 8 9 10
```

Submitted


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QUESTIONS
Q1
Print a hollow diamond pattern
5 marks
Q2
Design a class to calculate parking charges
5 marks
Q3
Print a number pattern with consecutive values
5 marks
Q4
Check whether a number is a spy number
5 marks
Q5
Print a butterfly pattern
5 marks
5/5 answered
Q4 Check whether a number is a spy number
5 marks
A number is a spy number if the sum of its digits equals the product of its digits.
TC_1:1124 → Spy number. TC_2:124 → Not a spy number. TC_3:-123 → Invalid input
Your Code
#include <iostream>
using namespace std;
int main() {
long long n;
while (cin >> n) {
if (n < 0) {
cout << "Invalid input" << endl;
continue;
}
long long temp = n;
long long sum = 0;
long long product = 1;
if (temp == 0) {
sum = 0;
product = 0;
} else {
while (temp > 0) {
int digit = temp % 10;
sum += digit;
product *= digit;
temp /= 10;
}
}
if (sum == product) {
cout << "Spy number" << endl;
} else {
cout << "Not a spy number" << endl;
}
}
return 0;
}
Submitted
Input
1124
124
-123
Output
Spy number
Not a spy number
Invalid input
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
5.00 / 5 marks
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QUESTIONS
Q1
Print a hollow diamond pattern
5 marks
Q2
Design a class to calculate parking charges
5 marks
Q3
Print a number pattern with consecutive values
5 marks
Q4
Check whether a number is a spy number
5 marks
Q5
Print a butterfly pattern
5 marks
5/5 answered
Q5 Print a butterfly pattern
5 marks
Print a butterfly pattern using nested loops.
* *
** **
*** ***
**** ****
****
**** ****
*** ***
** **
* *
Your Code
#include <iostream>
using namespace std;
int main() {
int n = 5;
for (int i = 1; i <= n; i++) {
for (int j = 1; j <= i; j++) cout << "*";
for (int j = 1; j <= 2 * (n - i); j++) cout << " ";
for (int j = 1; j <= i; j++) cout << "*";
cout << endl;
}
for (int i = n - 1; i >= 1; i--) {
for (int j = 1; j <= i; j++) cout << "*";
for (int j = 1; j <= 2 * (n - i); j++) cout << " ";
for (int j = 1; j <= i; j++) cout << "*";
cout << endl;
}
return 0;
}
Input
stdin input
Output